

7-day strength: response-surface model (30-run design)

Report: <https://claude.ai/artifact/Jj2co8WCAPA25XRNLemFcD> · Exploration: <https://claude.ai/artifact/EtGye3wqKVtkfaLToWL78i> Data: data/strength_7d.csv (Run 11 A/B = 0.42). The protocol was fixed before fitting, and the preliminary trends were tested only after the model and the optimum.

Protocol

Coded factors $-1..+1$: A = RCF (10–50), B = Carbonation (0, 0.5, 1, 5; numeric), C = A/B (0.42–0.48), D = SS (0–75). Box–Cox for the scale; sequential F + lack-of-fit for model order; backward elimination (α 0.10, hierarchy kept) cross-checked with the AICc-best subset; carbonation-coding check (switch only if pred R^2 +0.05 and AICc lower); bootstrap (2,000 residual + 2,000 case resamples of the whole selection) and leave-one-out stability; optimization with CI/PI and a leverage check.

Scale and order

- Box–Cox λ = 0.05, 95 % CI -0.45 to 0.60 , which excludes $1 \rightarrow \ln(\text{strength})$.
- On the $\ln$ scale: the quadratic adds significantly over 2FI (p = 0.0003), and lack of fit is not significant (p = 0.37).

Final model (pre-specified duration coding), $\ln$ scale

$\ln(Y) = 2.6960 + 0.2130 \cdot A + 0.0225 \cdot B + 0.4965 \cdot D + 0.1109 \cdot AB - 0.1206 \cdot AD - 0.1279 \cdot BD - 0.6087 \cdot D^2$
Actual units: $\ln(Y) = 1.1063 + 0.011140 \cdot \text{RCF} - 0.006391 \cdot \text{Carb} + 0.053937 \cdot \text{SS} + 0.0022179 \cdot \text{RCF} \cdot \text{Carb} - 0.00016085 \cdot \text{RCF} \cdot \text{SS} - 0.0013640 \cdot \text{Carb} \cdot \text{SS} - 0.00043283 \cdot \text{SS}^2$

- p-values: A 0.0001, B 0.59 (kept for hierarchy), $D < 0.0001$, AB 0.057, AD 0.034, BD 0.026, $D^2 < 0.0001$. A/B (C) terms were all removed.
- R^2 0.934 · adj 0.913 · pred 0.878 · adequate precision 18.6 · lack of fit p = 0.55 · SD 0.174 ($\ln$) · C.V. 7.7 %.
- Error predicting a new mix (leave-one-mix-out) is 2.25 MPa, against 2.16 MPa pure-error SD.
- The same 7 terms are the AICc-best and pred- R^2 -best of all 1,337 hierarchical subsets. The exact model was selected in 20/30 leave-one-out fits, and BD appears in all 30.
- Bootstrap selection (residual): A, D, D^2 100 %; B 94 %; BD 68 %; AD 60 %; AB 54 %; A/B terms 16–19 %.

- Diagnostics: no outliers (max $|t|$ 3.19, Run 14); Shapiro p 0.32; Breusch–Pagan p 0.47; Durbin–Watson 2.02; Run 27 has high leverage (0.75) but Cook's D is only 0.35.

Robustness

- On/off carbonation ($K = 1$ if carbonated) with the same 7 terms: adj 0.932, pred 0.913, AICc -94.9 vs -87.3 , lack of fit p 0.75, new-mix error 1.94 MPa. The duration terms among carbonated mixes are not significant (joint $p = 0.24$). On/off equation: $\ln(Y) = 1.0350 + 0.008749 \cdot \text{RCF} + 0.08307 \cdot K + 0.060996 \cdot \text{SS} + 0.008412 \cdot \text{RCF} \cdot K - 0.00015366 \cdot \text{RCF} \cdot \text{SS} - 0.009343 \cdot K \cdot \text{SS} - 0.00048369 \cdot \text{SS}^2$
- 4-level categorical carbonation is aliased (the 0.5 level has only 2 distinct mixes).
- On the raw MPa scale backward elimination keeps A, B, D, AB, D^2 (pred R^2 0.76). AD and BD drop (p 0.91, 0.18), so those two are partly scale effects.
- The 3-way term $\text{RCF} \times \text{Carb} \times \text{SS}$ is not significant ($p = 0.44$).

Best combination

- Carbonated RCF at RCF 50 %, SS ≈ 45 % (SS/SH ≈ 0.83), any A/B: 19.6 MPa predicted (on/off; 95 % CI 16.9–22.7) or 21.5 MPa (duration model at carbonation 5; CI 16.7–27.6). Leverage is within the design.
- Best uncarbonated: SS 53–62 % (SS/SH 1.1–1.6), 17.8–19.1 MPa; RCF matters little there.
- The carbonated and uncarbonated optima do not differ significantly ($p = 0.22$ duration; $p = 0.84$ on/off), so carbonation costs no strength at the optimum if RCF is high and SS ≈ 45 %.
- No run exists at the carbonated optimum. Suggested confirmation: triplicates at (carbonated, RCF 50, SS 45) and (uncarbonated, RCF 50, SS 55–60), plus a repeat of Run 25 (22.2 MPa, $1.3 \times$ its prediction).

Preliminary trends (tested after the model)

1. Higher RCF $\rightarrow$ higher strength: PARTLY. RCF 50/10 ratio for carbonated mixes is 1.46–1.99 up to SS 50 %; for uncarbonated mixes 1.42 at SS 0 but n.s. at SS ≥ 25 %. RCF never lowers strength significantly.
2. Carbonated weaker than uncarbonated: ONLY AT HIGH SS. The crossover SS is ≈ 18 % (RCF 10), 36 % (RCF 30) and 54 % (RCF 50); the largest loss is $\times 0.59$ at RCF 10 / SS 75 ($p = 0.0007$).
3. Longer carbonation $\rightarrow$ lower strength: NOT DETECTED. The joint duration test gives $p = 0.24$, and on/off coding fits better. Power is limited.
4. SS/SH $\geq 2 \rightarrow$ high strength; a lower ratio $\rightarrow$ lower strength: PARTLY. Strength peaks at SS/SH ≈ 1.2 – 1.6 (uncarbonated) and 0.8 – 1.1 (carbonated). At a ratio of 2, uncarbonated mixes are -3 % (n.s.) and carbonated -13 % ($p = 0.0007$). Below a ratio of about 1, strength falls steeply (peak $\rightarrow$ SS 0: -77 to -84 % uncarbonated, -63 to -73 % carbonated).

5. Carbonated better in NaOH-rich activators: SUPPORTED. At SS 0 the ratio is $\times 1.40$ at RCF 30 ($p = 0.003$) and $\times 1.65$ at RCF 50 ($p = 0.0007$), n.s. at RCF 10. Carb $\times$ SS $p = 0.0003$ (on/off).

Assumption: $SS/SH = SS/(100 - SS)$, with SS (%) taken as the sodium silicate share of an SS + NaOH-solution activator.